

Novel High-Efficiency Broadband Microwave Rectifier Design Based on Adaptive Harmonic Control

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Abstract—Conventional class F and inverse F microwave rectifiers achieve high efficiencies by harmonic control. However, the working bandwidth is limited since the harmonics generated during rectification have to be strictly controlled. In this paper, the microwave rectifier adaptively switches between class F and inverse class F when the operating frequency varies by a harmonic control network. It ensures the rectifier has high conversion efficiency and a broad frequency range. A broadband rectifier is realized with an efficiency greater than 70% from 1.55 to 2.65 GHz at 11 dBm input power. The design and measured results of the rectifier verify the proposed adaptive theory on harmonic control to achieve high rectifying efficiency, which successfully breaks the frequency limitations of harmonic terminal control.

Index Terms—broadband, energy harvesting, harmonic control, rectifier, wireless power transfer.

I. INTRODUCTION

Microwave Wireless Power Transfer (MWPT) and Energy Harvesting (EH) technologies have demonstrated significant research importance and broad application prospects across various fields[1][2]. They enhance the convenience and safety of equipment and reduce dependency on traditional cables and batteries, contributing to environmental protection. A highly efficient and stable rectifier is essential for converting the received electromagnetic energy. The rectifier must efficiently convert energy across different frequencies, posing a challenge for its design. Our research addresses this challenge and presents a practical solution with broad application potential.

The efficiency of the rectifier can be significantly improved through proper harmonic termination design.[3] utilized a series band-stop structure to achieve harmonic recovery, achieving an efficiency of 80.9% at 2.45 GHz. [4] derives an analytical model to calculate the rectifier efficiency of class F rectifiers based on the diode voltage and current waveforms in the time domain. A rectifier designed by the continuous F-class concept only achieves efficiencies greater than 50% within the range of 0.78 GHz to 1.43 GHz [5]. Designing broadband matching networks [6][7] and impedance compression networks is an effective design method to expand bandwidth further. The non-uniform transmission lines have been employed for impedance matching, successfully achieving high rectification efficiency over a wide frequency range [8]. Impedance compression techniques can achieve compact size and high efficiency by reducing the variation range of the rectifier's input impedance [10][11][12].

The strict frequency requirements limit the application of harmonic termination techniques in wideband rectifiers. A

major challenge is to expand the operating bandwidth with high efficiency. Therefore, this paper proposes a high-efficiency wideband rectifier design scheme based on a novel harmonic control network (HCN). This HCN allows the rectifier's harmonic termination mode to vary with frequency, enabling the design to switch from a standard inverse class F to class F. This ensures high efficiency through harmonic termination control while expanding the rectifier's operating bandwidth.

II. HARMONIC-TUNED RECTIFIER THEORY

In an inverse Class-F rectifier, the fundamental frequency voltage V across the diode can be expressed using the following equation:

$$V(\theta) = \begin{cases} 2V(f_0)\sin\theta, & 0 \leq \theta < \pi \\ 0, & \pi \leq \theta < 2\pi. \end{cases} \quad (1)$$

The Fourier expansion of the DC voltage of the waveform is represented by:

$$V_{DC} = \frac{2V(f_0)}{\pi} \quad (2)$$

According to the waveform distribution, the current is given by the following equation:

$$i(\theta) = \begin{cases} 0, & 0 \leq \theta < \pi \\ 2I_{DC}, & \pi \leq \theta < 2\pi. \end{cases} \quad (3)$$

The Fourier expansion of the current waveform for dc component is

$$I_{DC} = \frac{\pi}{4} I(f_0) \quad (4)$$

DC loads are thus derived

$$R_{DC} = \frac{8V(f_0)}{\pi^2 I(f_0)} = \frac{8}{\pi^2} R(f_0) \quad (5)$$

The conversion efficiency in the ideal case can be calculated as

$$\eta = \frac{P_{DC}}{P(f_0)} = \frac{2V_{DC}I_{DC}}{V(f_0)I(f_0)} = \frac{2 \frac{2}{\pi} V(f_0) \frac{\pi}{4} I(f_0)}{V(f_0)I(f_0)} = 1 \quad (6)$$

To verify the rectifier's performance under different operating modes, the ADS (Advanced Design System) software was used for voltage and current waveform simulations under large-signal conditions. Fig. 4(a) shows the voltage and current waveforms at $f_1=1.6$ GHz, which closely resemble a standard inverse Class-F waveform. Fig. 4(b) illustrates the waveforms at $f_2=2.5$ GHz, which approximate those of the ideal Class-F operation mode.

To verify the feasibility of the design theory, a rectifier based on the proposed design was fabricated. The rectifiers were implemented on an F4B-M dielectric substrate, with a relative permittivity (ϵ_r) of 2.65 and a loss tangent ($\tan \delta$) of 0.0012. The substrate thickness is 1 mm. Fig. 5 shows the layout and physical prototype of the broadband rectifier. The specific design parameters are also detailed in Fig. 5.

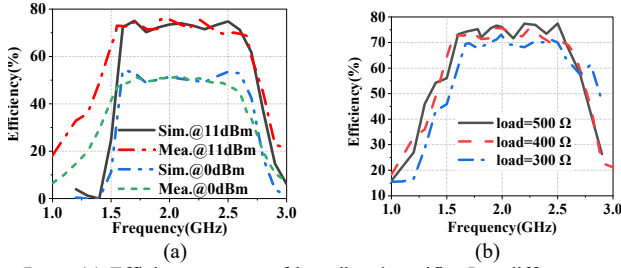


Fig. 7. (a) Efficiency curves of broadband rectifier I at different powers. (b) Efficiency curves of broadband rectifier I at different dc loads.

TABLE II

Comparison between BROADBAND Rectifiers

Ref	PCE >70 % (GHz)	BW (%)	Power (dBm)	PCE > 40% at 0 dBm (GHz)	BW (%)	Size (mm ²)
[8]	2.0-3.05	41.5	10	1.9-3.1	48	36×35
[6]	2.08-2.58	21.5	15.5	2.03-2.75	30	126×68
[7]	0.57-0.90	44.9	12.8	N/A	N/A	60×88
[11]	2.1-3.3	44.4	14	NA	NA	31×18
This Work	1.54-2.66	53.3	11	1.52-2.76	57.9	32×25

Fig. 6 shows the measurement results of S-parameters. Fig. 6. The simulation and measurement results align closely. Under RF power levels of 0 dBm and 11 dBm, $|S_{11}|$ of rectifier I remains below -10 dB across the frequency range of 1.5 GHz to 2.7 GHz, indicating good impedance matching.

Then, the power conversion efficiency (PCE) of rectifier I was measured. The simulation and measurement results of rectification efficiency under different input power levels are shown in Fig. 7(a), with a DC load of 400 Ω . The simulated efficiency closely matches the measured results. Under an RF input power of 11 dBm, the rectification efficiency exceeds 70% across the frequency band of 1.54 GHz to 2.66 GHz, achieving a fractional bandwidth of 53.3%. When the input power decreases to 0 dBm, the rectification efficiency remains near 50%. Measurement results for different output loads are shown in Fig. 7(b).

Table II compares the broadband rectifier designed in this work with some previous designs. The designed broadband rectifier has a significantly larger operating bandwidth than other designs. It validates the advancement and effectiveness

of our design theory, considerably extending the operational range of the rectifier

IV. CONCLUSION

This paper presents a design method for ultra-wideband rectifiers based on frequency-adaptive harmonic control, which improves the rectifier's efficiency and extends its bandwidth. The paper begins with an analysis of how harmonic terminal control can enhance conversion efficiency, followed by an examination of the working principle of the harmonic control network. A broadband rectifier was fabricated and tested to validate the theory's feasibility. The high efficiency and wide working bandwidth demonstrate the advances and effectiveness of the proposed theory.

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